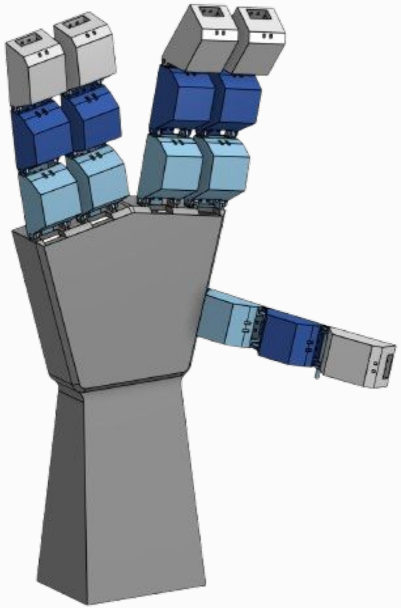
Manus



Voice-Controlled Robotic Hand

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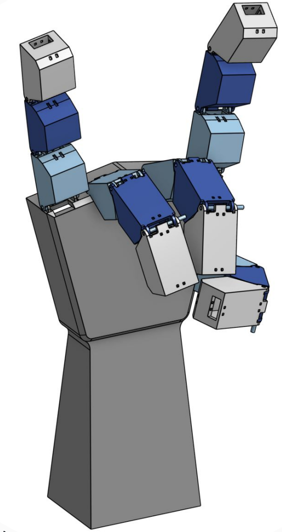
01

Project Overview



01

Need for project: over 5 million Americans have upper limb loss or impairment that limits their ability to perform basic grasping tasks. Idea is to design a voice controlled prosthetic arm that allows users with little to no finger mobility to execute grip patterns

02

Current examples: many examples exist online, which we borrowed inspiration from. In most cases, they are actuated using servos connected to fishing lines that pull a set of jointed fingers. Our project is called “Manus” or Latin for hand

03

Theoretical/conceptual problem solving: it is a prototype arm extension for people who do not have hands. Voice recognition exists but it can also be manually controlled using a website

Sensing, Actuation, & Data Transmission

01: Sensors

Voice control module controls servos by recognizing audio signals and mapping it to commands.

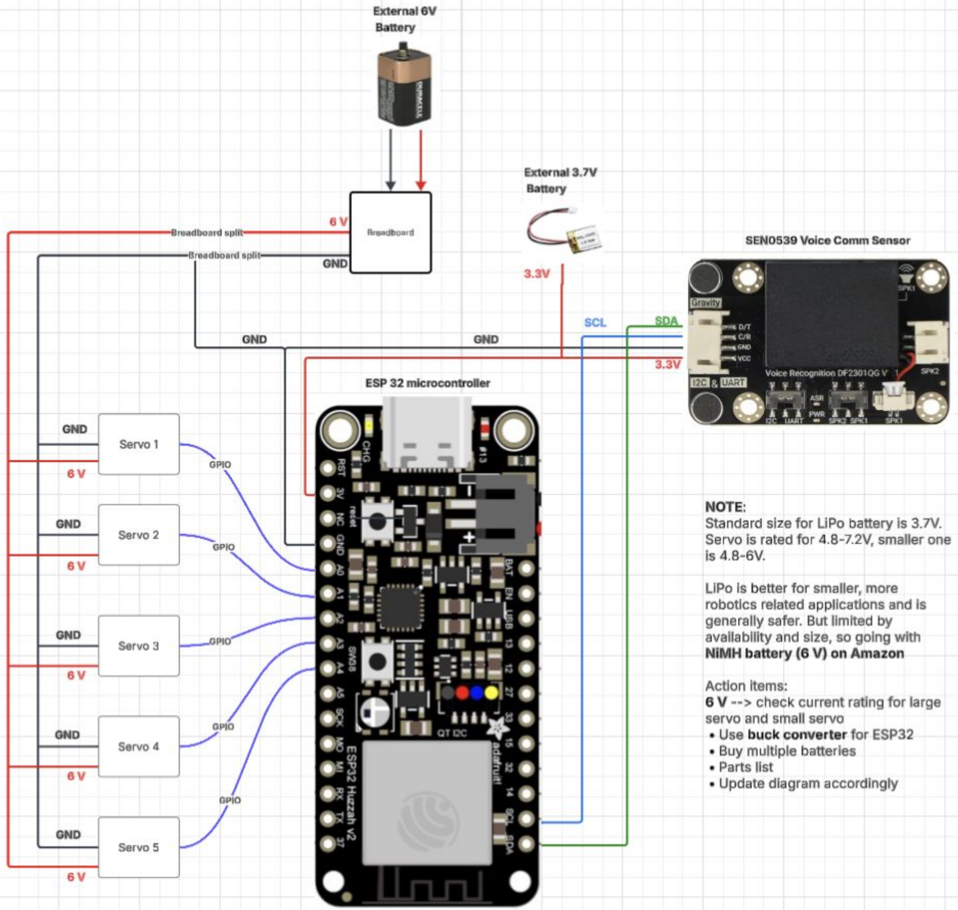
Servos have potentiometers which give us feedback on angle and allow us to manually move fingers to various positions.

02: Actuation

Fingers actuate based on verbal commands and manual movement. Fingers are controlled by servos that pull wire in the strings till they are fully contracted. They release tension allowing for fingers to be extended again.

03: Data Transmission

Mqtt for sliders to change servo angles, state of hand shown



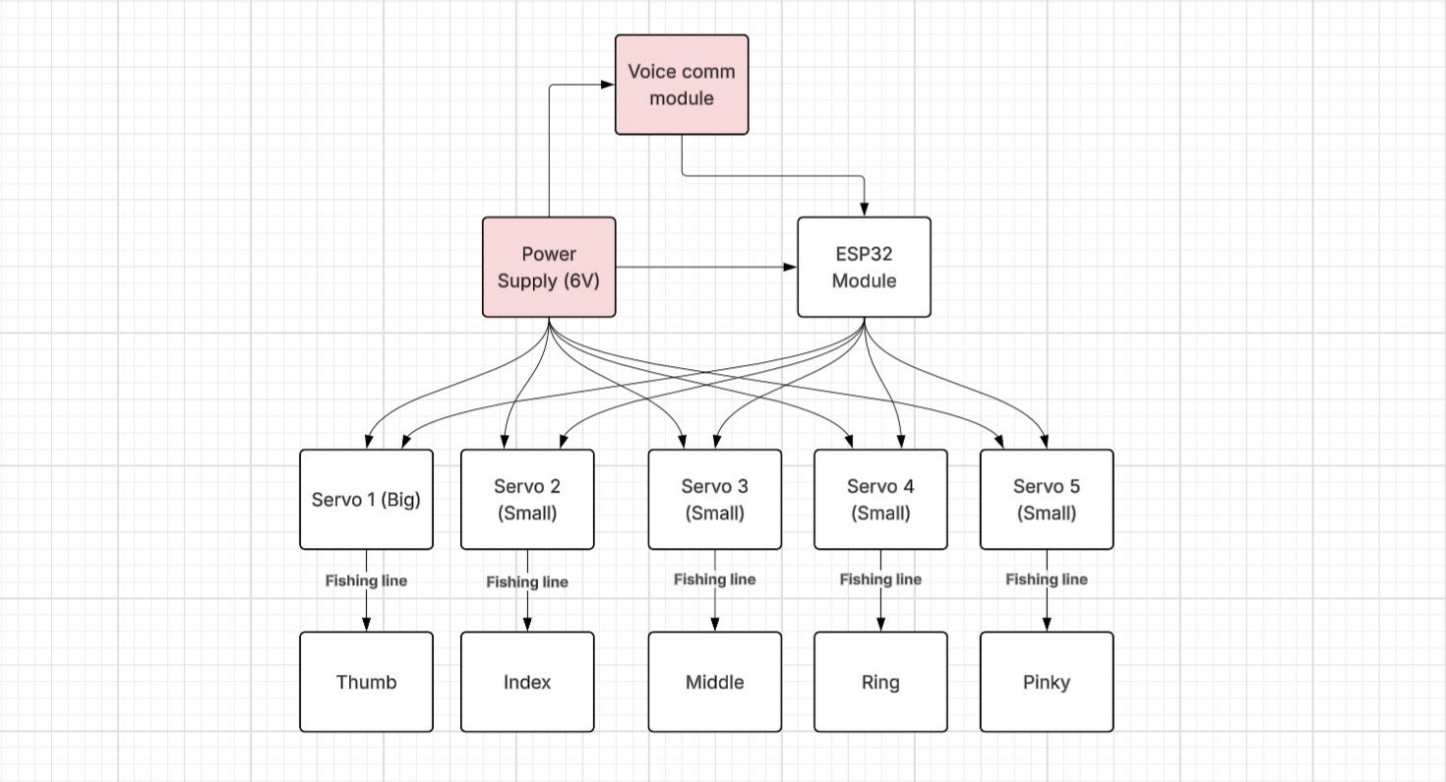
Wiring Schematic

The ESP32 uses I2C to communicate with the voice comm module. Both are connected to a 3.7V LiPo battery.

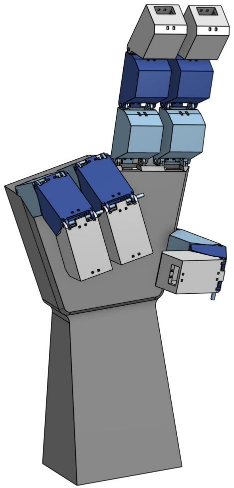
The servos are connected to a separate 6V NiMh battery to prevent sudden current surges from servos on the ESP32 power line. All of their sensing wires connect to a GPIO pin on the ESP for PWM signals to the servos and angles to the potentiometers.

Everything shares a common ground.

Wiring Schematic (continued)



03 Challenges



01

Challenge 1:

Mechanical linkages were difficult to model and prone to breaking, greatly increasing complexity for assembly and testing. Forced us to make a lot of on the fly decisions to have a working prototype (printed hardstop, groove changes)

02

Challenge 2:

String displacement for each finger was difficult to calculate, so required a lot of calibration time. Motor torque at times was too high, breaking certain parts

03

Challenge 3:

Testing with separate battery was difficult - circuit issues led to the voice control module being inconsistent at times. Connected all to the same ground and ensured that 6V battery was exclusive to servo and ESP was exclusive to voice comm

Live Demo

Thank You